



Global Quantum + AI Challenge 2026

Enterprise Challenge Statement

1. Challenge Title

Quantum-Enabled Grid Expansion Planning for Distribution System Energy Networks

2. Executive summary

The energy transition is driving rapid growth in renewable electricity and a steady decline in fossil fuel-based generation. Grid infrastructure now has to support more variable and distributed power flows while at the same time dealing with increased energy demand and keeping the network resilient and secure. To accommodate this, energy grid operators are looking to expand and reinforce the medium and low voltage networks while respecting operational constraints like voltage limits and thermal ratings. The network expansion problem is an optimization problem which looks at trade-offs for grid operators that want low investment cost while also reducing technical losses, curtailment, and outage risk. This creates a large combinatorial design space with build or no build decisions tightly coupled to network physics across many plausible operating scenarios. In this challenge, teams will apply quantum enhanced methods to classically hard benchmark instances of the network expansion problem in order to benchmark and potentially demonstrate evidence of practical quantum advantage.

3. Industrial and Operational Context

Distribution System Operator (DSO) planners forecast load growth and distributed generation uptake, build scenario sets for future operating conditions, then assess feeders, transformers, and substations against limits using power flow studies and engineering standards. Key stakeholders include DSO planning and asset management teams, operations and protection engineers, and downstream municipalities and permitting bodies. Today this is done via grid analysis tools for load flow and contingency checks, with much of the option screening, cost estimation, and reporting still done in spreadsheets and engineer in the loop analysis. Candidate grid line reinforcements are often generated by rules and experience, then iterated through simulation, with practical constraints such as permitting timelines, outage windows, and constructability decide what is feasible. The main bottlenecks are the combinatorial explosion of build or no build choices across many scenarios and therefore choose a potentially more conservative plan as they cannot evaluate more solutions.

4. Challenge Objective

4.1 Primary objective

The overall objective of this challenge is to develop a quantum solution to solve the grid expansion problem that optimally reduces network congestion i.e. minimize the congestion amount per line. This problem can be posed as a combinatorial optimization problem. As a framework for fairly comparing potential quantum advantage in optimization problems, we follow the approach laid out in [5] in this challenge. Namely, first identifying hard problem instances of a (potentially simplified) problem formulation and then using these benchmark instances and resulting problem formulation to shape the quantum algorithmic approach.

As a primary objective and foundation for this challenge, a variety of problem instances should be identified that scale poorly (e.g. exhibit exponential or high degree polynomial scaling), or cannot be proved to be optimal within a reasonable amount of runtime using traditional classical solvers like Gurobi or CPLEX. See for example [5] for other related graph theoretic problems such as network design, Steiner tree packing and topology problems.

The objective function and constraints may vary mathematically to determine these hard instances as long as the end formulation is a variant of those used in network expansion literature. The problem instances should ideally be hard for small problem instances and are suitable for running with quantum optimization on real quantum hardware (NISQ). The following are a few preliminary problem formulations to help with the challenge, but we encourage the team to research further into alternative formulations that balance between the business objective, have a quantum approach to solve, and are hard classically. In [1], the problem is formulated as a binary decision problem of whether to include a powerline subject to several constraints and aiming to minimize the overall cost. In [2], the problem is formulated as a minimum capacitated Steiner tree problem where minimum paths are identified to meet energy demand and reduce the cost of adding power lines. In [3], the related network reconfiguration problem is studied and proven to be strongly np-hard. In [4], the authors propose a Binary Linear Programming model for offering suggestions of line additions to a network that reduce the number of grid congestions.

A quantum algorithm shall be developed that can run and benchmark at least one of the hard problem instances. Note this can be variational in nature (QAOA, VQE based), or

quantum machine learning based, or any other use of quantum computers in a hybrid fashion to work together with classical compute.

4.2 Secondary objectives

The identified hard problem instances shall have differing numbers of variables. At least one of these hard problem instances will be small enough to run using hybrid quantum methods on current NISQ hardware (subject to circuit depth constraints/limitations). As a secondary objective, the goal would be to show that for a given problem instance, running it with simulator (e.g. MPS) gives poor performance. This is to show that the overall resulting objective value is poor and therefore acts as a motivation that running on quantum hardware could provide a potential quantum advantage. The quantum approach should aim to be scalable in the sense that it works for low number of qubits (which implies grid sizes) up to utility-scale (>100 qubits).

5. Problem Definition

5.1 Inputs

We have provided 4 starting references for the scientific literature survey and explained that the exact problem formulation with regards to the optimization cost function can be left open for simplification. The discipline of distribution grid expansion, and more generally, transmission grid design, is a broad area of electrical engineering research. The key component is that the challenge is about the distribution grid, and, the end goal is to optimally increase an initial grid size in order to reduce grid congestions (reduce the number of line flow violations).

5.2 Outputs

The outputs of the optimization should ultimately be a set of new power lines to be added to an existing electrical grid, as well as the underlying cost function values and reduction in congestions (or near congestions). The precise metrics will depend on the exact problem formulations and is up to the participants. Ideally, at the end of the challenge, E.ON will like to collaborate with the winning team to finalize any results and publish a scientific paper together.

5.3 Constraints

Submissions are recommended to use an open-source quantum framework (e.g., Qiskit, PennyLane, or Cirq). Submissions should include a compatibility layer enabling the E.ON team to execute and validate results in qiskit upon project completion. The developed solution should run within a runtime limit of a few hours

5.4 Scope and assumptions

The data used for the problem formulations is open source.

6. Data and Resources

The scientific literature references below contain links to open-source datasets such as IEEE test network cases. Synthetic grids may be required to be generated that have the same graph properties as these networks. At later stages of the project, E.ON will make available example subgrid networks from the medium and low voltage grids which have been anonymized. Here are some additional useful data and resources:

1. IEEE Test Networks:
https://pandapower.readthedocs.io/en/latest/networks/power_system_test_cases.htmlhttps://pandapower.readthedocs.io/en/latest/networks/power_system_test_cases.html
2. California Test System:
<https://arxiv.org/abs/2210.04351><https://arxiv.org/abs/2210.04351> and repository <https://github.com/WISPO-POP/CATS-CaliforniaTestSystem><https://github.com/WISPO-POP/CATS-CaliforniaTestSystem>
3. Texas A&M Test Case Repository: <https://electricgrids.engr.tamu.edu/electric-grid-test-cases/><https://electricgrids.engr.tamu.edu/electric-grid-test-cases/>
4. QOBLIB (see Network Design, Steiner Tree, and Topology which are related datasets): <https://git.zib.de/qopt/qoblib-quantum-optimization-benchmarking-library>

7. References

- [1] Gust, Gunther, et al. "Designing electricity distribution networks: The impact of demand coincidence." European Journal of Operational Research 315.1 (2024): 271-288.

- [2] Duan, Gang, and Yixin Yu. "Power distribution system optimization by an algorithm for capacitated Steiner tree problems with complex-flows and arbitrary cost functions." *International journal of electrical power & energy systems* 25.7 (2003): 515-523.
- [3] Khodabakhsh, Ali, et al. "A submodular approach for electricity distribution network reconfiguration." *arXiv preprint arXiv:1711.03517* (2017).
- [4] Cuenca, Juan J., et al. "Event-informed identification and allocation of distribution network planning candidates with influence scores and binary linear programming." *IEEE Transactions on Power Systems* 40.1 (2024): 492-504.
- [5] Koch, Thorsten, et al. "Quantum Optimization Benchmark Library--The Intractable Decathlon." *arXiv e-prints* (2025): arXiv-2504.
- [6] Kotil, Ayse, et al. "Quantum approximate multi-objective optimization." *Nature Computational Science* (2025): 1-10.